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Public for private: the relationship between public and private school enrollment in the Philippines

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Abstract

To overcome market failures due to externalities or lack of an effective credit market, countries often try to improve access to education by providing more public school places. However, if there is already an active private sector, then some of the public expansion may draw away students who may have gone to school anyway. This would attenuate any efficiency gains. This paper estimates this crowding-out effect for the Philippines, a country with a large private education sector. Using regional data over the past 10 years, we estimate that the large expansion in the public secondary education sector is negatively associated with private secondary enrollment. The range of the response is around four or five fewer private school students for an increase of 10 public school students. On the other hand, the crowding-out effect is insignificant at the primary and tertiary levels. © 2001 Elsevier Science Ltd. All rights reserved.

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1. Introduction

In many countries, the private sector is an important provider of education. In countries as diverse as Australia, Belgium, Botswana, Chile, Comoros, Gabon, Korea, Lebanon, Madagascar, Mauritius, the Netherlands, the Philippines, and Tanzania, over 30% of secondary school students are in private school.¹ The performance of the private sector can thus have a key role in explaining national educational outcomes. Particularly in developing countries, they offer an alternative to edu-

cational expansion when government budgets and capacity to deliver are severely constrained.

Governments can affect the size of the private sector by direct regulation. Some go so far as to prohibit private schools.² Others impose less restrictive regulations which nonetheless can limit the participation of demanders or behavior of private suppliers, such as what fees private providers can charge or skill requirements of teachers.

But often, direct regulation is not the main way that governments affect private outcomes. Regulations may not exist, or, if they do, lax enforcement renders many of them ineffective. Even in these settings the public sector can greatly influence the private sector. The provision of public education in itself can affect private enrollment by drawing away demand. For example, heavily subsid-

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¹ See van der Gaag (1995) and World Bank (1995) for some recent figures on private–public spending. Public–private enrollment figures are available from The World Bank (2000).

² As Pakistan did in the 1970s; see Jimenez and Tan (1987).

ized public schools may be attractive alternatives even if they are of lower quality than private schools.

The sensitivity of private to public enrollment is important for evaluating public policy. Most providers of public schools assume that increasing public school enrollment has a direct and immediate effect on overall enrollment. In fact, this relationship is most often considered to be one-for-one. But this would not hold if students were simply attracted to the public sector from another type of school. In this case, the *net* gain in enrollment may be significantly less than the amount of the public sector expansion. In evaluating the effect of policy, it would then be important to look beyond the public sector.

In countries where the private sector is extremely small or caters to a small elite, marginal changes would not matter. But where the private sector is active and provides a viable alternative to the public sector, substitutability between private and public sectors is important in evaluating the net benefits of government expansion. While government interventions in education are usually justified when there are market failures due to externalities or the lack of an active credit market, this substitutability would impede overall improvements of school enrollments. If the substitution effect is strong, the efficiency gain from government interventions may be attenuated significantly.

This is exactly the case in the Philippines, where the private sector is an important provider of education. About 22% of all students are in the private sector. The proportion varies across levels and ranges from a relatively small 7% in elementary, to 31% in secondary and 77% in tertiary (see Fig. 1). The Philippines is also a potentially good case study because there have recently been dramatic changes in the amount of public

enrollment. This is especially true at secondary levels, the focus of this study, where the share of private education has fallen from 46% in 1982 to 31% in 1997 (Fig. 1).

What is the magnitude of the substitution effect? We investigate this issue by using a newly available regional data set. In Section 2, we describe the institutional context of private education in the Philippines and use it to motivate a simple empirical framework, described in Section 3, to measure the substitution effects between public and private education. Section 3 also describes the data to be used. We analyze the empirical results in Section 4 and discuss briefly some policy implications in Section 5.

2. Institutional background

Stimulated by a high population growth rate and a traditionally high school participation rate, enrollment in both private and public schools has been rising at all levels in the Philippines (World Bank and Asian Development Bank, 1999). This has undoubtedly been partly fueled by a rapid expansion of places in public schools. Between 1986 and 1994, for example, total public spending on education rose from 2% to 3% of gross domestic product (GDP). The question is to what extent the large public sector expansion reflects a net gain in enrollment for Philippine education, by attracting those students who may not have continued on with their schooling, or simply a substitution between private and public schooling.

Public education in the Philippines has a fairly recent history, dating back to the early part of the 20th century. Before then, students attended religious as well as non-sectarian private schools. This tradition of a large private role in provision continued even after the establishment of the public school system and still persists today, particularly at higher levels. There are large differences between quantity, quality and unit costs between private and public institutions at various levels, and even among the institutions. At the primary level, the small private sector tends to provide higher-quality schools for the relatively small proportion of households who have opted out of the public system. At the secondary and tertiary levels, private school quality spans a wider range, from some of the very best to some of the very worst. These institutions include non-profit schools run by religious groups or non-sectarian foundations, as well as profit-making entities that produce a substantial rate of return to their owners.

In 1997 about 21% of the country's 16 million students were in private institutions. Less than a tenth of those in primary schools were private school students, as compared to about a third for those in secondary and over three-quarters at tertiary, including both traditional

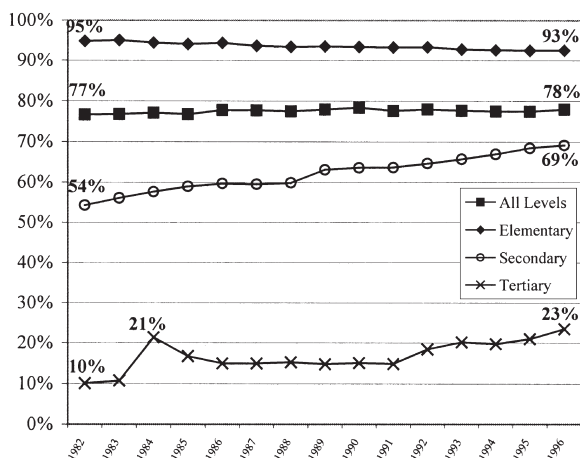


Fig. 1. Public sector enrollments as percentage of total enrollments, 1982–96. Source: World Bank and Asian Development Bank (1999).

higher education and vocational/technical schools (Fig. 1). Compared with the rest of Asia, the relative size of the private sector in the Philippines is below average for primary and above average for secondary and tertiary.³

Fig. 1 also shows that there have been significant changes in the private–public shares of enrollment since the early 1980s. The share of public enrollment has risen 13 and 15 percentage points in tertiary and secondary from levels of only 10% and 54% in 1982, respectively, while that in public elementary has dropped by two percentage points. The policy question is: how much of the variation in total enrollment rates can be explained by these relative shifts in shares? In particular, has the increasing share of public come partly at the expense of the private sector so that the net enrollment gains cannot be fully attributed to public action? The answers may vary depending on the level of education.

If government schools provided either such high quality or low prices that the number of students who wanted to enter exceeded the number of places the government is willing or able to provide — due, for example, to budgetary restraints — there would be excess demand for public school places. In the Philippines, this is arguably the case for at least a part of the tertiary educational system. The University of the Philippines' main campus, its school of agriculture and perhaps a few other campuses are high-quality institutions that offer highly subsidized education for those who are able to succeed in the merit-based entrance examinations. In such a setting, an increase in public school places would lead to an immediate increase in public enrollment. However, the effect on the private sector would depend on whether the additional enrollment is attracted from the private system or from the youth who would otherwise not have sought higher education. It is not possible to predict *a priori* what the effect would be.

In secondary education, the situation has been evolving. The private sector made up almost half of enrollment as recently as the early 1980s and offered a wide array of educational choices, from low to high quality. Before that, the situation was similar to that in universities. Now, the private and public systems implicitly compete for students by adjusting prices and quality. An increase in public enrollment in this setting would probably draw students from the private sector if it provided similar quality at lower prices.

In primary education, the public sector provides free mass education for over 90% of the population. The private sector serves only a “niche” market of high-quality, high-priced schools. In this case, an expansion of public sector enrollment and upgrade of quality could draw students from the private sector only if it encroached on the private niche. This is unlikely to be a strong effect in

the Philippines, where the public sector is relatively homogeneous.⁴

To subject these hypotheses to the empirical evidence, we develop in the next section an estimation model for the demand for private education in the Philippine context.

3. Empirical framework

3.1. Deriving the estimating equation

For any given level of education, Philippine households and students can choose between public and private options. In the Philippines, students are not restricted to attend the public school that is closest to them since all public education is funded centrally rather than locally. They can also choose not to continue with their schooling; although school participation rates are high in the Philippines, relative to developing countries, so are dropout rates. Students or their parents will choose an option which gives them the highest level of well-being — for example, an option which gives them the highest expected increment in future income relative to the cost they have to incur.

We develop an intertemporal framework to analyze the investment demand for education over a household's life-cycle (Levhari & Weiss, 1974).⁵ A household will determine the optimal level of schooling so as to equalize the net marginal productivity of schooling and the cut-off interest rate for investments. Under perfect capital markets, household income would not affect this demand. However, if capital markets are imperfect, the cut-off interest rate is endogenously determined as a negative function of income, and schooling will be positively related to income.⁶ Studies show this to be the case for many developing countries (for example, see Behrman & Knowles, 1999).

⁴ See James (1987, 1993) for a discussion of differentiated demand and its effect on the demand for private education.

⁵ Alternatively, in a consumption framework, households compare the indirect utility from public schooling, V_g , and private schooling, V_p , based on the relative prices and quality of the two options plus socio-economic characteristics. If $V_g > V_p$, the household will choose public schooling. One can derive a standard demand equation for private education from this framework. As with normal private goods, demand for private education will be a negative function of own prices. Demand will also be a positive function of income. This could be because schooling is a normal consumption good.

⁶ We are grateful to an anonymous referee for pointing this out.

³ See World Bank (2000).

This investment story is illustrated in Fig. 2. The horizontal line at $r(y)$ indicates the cut-off rate of educational investment, which is a negative function of household income, y , under the assumption of capital market imperfection. The downward sloping lines represent the net value marginal productivity of schooling, $NVMP^{Pub}$ for public schools and $NVMP^{Priv}$ for private schools. Each $NVMP$ curve is affected negatively by price or cost of schooling, p , and positively by school quality, x , $NVMP = NVMP(p, x)$. If, as shown in Fig. 2, $NVMP^{Pub}$ is steeper than $NVMP^{Priv}$, public schools are more productive than private ones at lower levels education. (Of course, the other relationship is also possible.)

A household selects the school type which maximizes the returns to education (or along the bold lines in Fig. 2). The demand for schooling, q , is optimally determined at the intersection of $r(y)$ and $NVMP(p, x)$. If $r(y) = NVMP^{Priv}(p_p, x_p) > NVMP^{Pub}(p_g, x_g)$, private school is chosen (p_p and x_p denote, respectively, the price and quality of private education, while p_g and x_g denote that of public education). If $r(y) = NVMP^{Pub}(p_g, x_g) > NVMP^{Priv}(p_p, x_p)$, public school is chosen.

Thus, the numbers of students in private and public schools, n_{priv} and n_{pub} , respectively, are functions of all price and quality variables, i.e., p_p, x_p, p_g, x_g as well as aggregate income (Y). Moreover, the total number of school-age children (N) is entered as an independent variable as well, capturing the cohort-specific scale effects on enrollment. This paper uses this framework to postulate a simplified estimating model of enrollment in private education for region j at time t as:

$$n_{priv,jt} = \alpha_0 + \alpha_1 p_{p,jt} + \alpha_2 x_{p,jt} + \alpha_3 p_{g,jt} + \alpha_4 n_{pub,jt} + \alpha_5 x_{g,jt} + \alpha_6 N_{jt} + Y_{jt} \beta + \alpha_j + \alpha_t + u_{jt}, \quad (1)$$

where u is an i.i.d. mean zero error term. This paper will

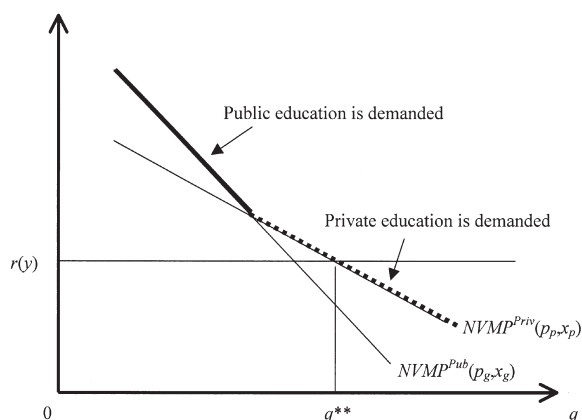


Fig. 2. Illustration of investment model of a household's demand for schooling. The case of two school types, private and public.

estimate the above equation using data at the provincial level over time in the Philippines from 1986 to 1997.

What are the expected signs of the variables? The coefficients α_1 and α_2 reflect, respectively, the own price (negative) and quality (positive) effects within the private sector.

The coefficients α_3 , α_4 and α_5 capture the substitution effects with respect to the public sector. If private and public education are substitutes, then, in a market setting, the price in the public sector should be positively related to demand in the private sector — we would expect that $\alpha_3 > 0$. This is the classical measure of substitution and the first of three used in this paper.

However, it is unlikely that prices alone will reflect the relative scarcity of public sector places. Prices are set independently of demand in the public sector. They are set at a relatively low and uniform rate. For example, in public secondary schools the average tuition is 165.3 pesos with a standard deviation of 44.0, while in private secondary schools, the average tuition is 1596.8 pesos with a standard deviation of 252.9. It is thus important to include a more direct measure of the availability of public school places — such as enrollment in the public sector,⁷ since public school places may be rationed. A second indicator of substitution would then be that private enrollment is negatively associated with public enrollment or $\alpha_4 < 0$.

The third type of substitution effect is that the quality of education in the public sector is negatively associated with the quantity of private education, or $\alpha_5 < 0$. Higher public quality will attract students away from the private sector unless there is excess demand for education overall, which, as discussed earlier, is more likely to be the case for tertiary education.

Income would have a positive effect on demand since it would alleviate the capital constraint. However, the positive effect of income may be mitigated if the measure of income such as the average wage in the community is also a proxy for the earnings that a student may earn had he or she been working. Income earning opportunities may induce dropout. Hence, the sign of β may be ambiguous.

The terms α_j and α_t represent region- and time-specific fixed effects, respectively.

3.2. Estimation issues

Estimating and interpreting Eq. (1) raises some important econometric issues. First, in order to interpret the estimated relationship as demand, one must parameterize supply. We are not able to estimate a demand

⁷ It is also important to get other measures of substitutability because prices of quasi-private goods such as education are measured so imprecisely.

and a supply curve simultaneously. However, we can estimate Eq. (1) as single-equation model and still interpret it as a demand curve if we assume that the long-run private sector supply curve is perfectly elastic. This would be so if there are constant returns to scale. Otherwise, one could not hold constant for supply parameters and the estimated relationship would have to be interpreted as a reduced-form relationship.

Second, for the initial estimation, we assume that public sector enrollment is determined independently of the price, size and quality of the private sector. This can be justified because decisions about public education are often made without regard to their effect on the private sector.⁸ For example, the government has interpreted the constitutional stipulation to provide secondary schooling for all Filipinos by doing away with public school fees. But even if this were the case, public enrollment could still be affected by private enrollment for the same reasons that private is affected by public enrollment. Households and students carefully weigh the pros and cons of each alternative and it is possible that private prices and quality affect public enrollment. Moreover, if any private school places are rationed, changes in that rationing device could affect public enrollment, even after holding constant for prices and quality.

In order to account for this possible endogeneity of the explanatory variable, we use the instrumental variable method for public sector enrollment.⁹ As is always the case in this technique, the challenge is to find an appropriate identifying instrument — one that is correlated with public sector enrollment but not with private sector enrollment. One such instrument is an election year dummy that captures a regime change in the public sector.¹⁰ The data set used in the analysis is for regions in the country from 1987 to 1998. The 1992 elections ushered in the government of President Fidel Ramos. Such changes can be expected to have a direct link to public

sector spending but affect the private sector only indirectly, since the regulatory environment is weak. In fact, the correlation between the 1992 regime change variable and public enrollment is 0.25 while that between the same variable and private enrollment is -0.06 .

3.3. Data

The analysis is conducted at the regional level, for which annual data are available from 1986 to 1998. There are 13 planning regions in the Philippines, plus the National Capital Region (Metro Manila), and two special regions that constitute politically autonomous areas in Muslim Mindanao and northern Luzon. The data have been compiled from various existing sources by the Upecon Foundation, a non-profit organization housed within the School of Economics of the University of the Philippines.

The dependent variable is private sector enrollment, as conditioned by public enrollment. Enrollment data for private and public education for primary, secondary and tertiary levels are available from the Department of Education, Culture and Sports (DECS). Data for some of the specialized regions are unavailable for some years and these observations are not included in the analysis. We could of course have done the analysis using enrollment rates but that would have overly restricted the specification — we instead use enrollment and use the number of children in the relevant age category as a control variable.

The quality of education is proxied by scores in standardized tests. The best available measures are standardized tests in 6th grade, the National Educational Assessment Test (NEAT), and for the secondary level, the National Scholastic Aptitude Tests (NSAT) in 10th grade. While NEAT test scores are available since 1994, detailed NSAT scores are available only from 1995 on. The tests were given on a sample basis to selected students in all sectors until the recent regulation change which made these tests mandatory for all students.

These tests do not capture all the dimensions of quality that society may value. Other outcome measures that might capture other important aspects of quality are cohort survival rates and dropout rates. The problem is even more difficult for the post-secondary level where there are no standardized tests for the wide range of subject areas. The best proxy would be performance in professional licensure examinations, which leave out many disciplines that do not have such examinations and many pupils who choose not to take them. These limitations must be borne in mind in interpreting the results.

The “prices” for primary and secondary education are obtained by surveys conducted by the Foundation for Assistance to Private Education (FAPE). They include the average costs incurred by families for each province and year for the following: school fees (tuition) and other

⁸ In the Philippines, education is financed by government revenues. Decisions about resource allocations are made centrally, in consultation with the centrally appointed regional directors (World Bank and Asian Development Bank, 1999). This is in contrast to the situation in US schools, where per pupil public school spending across localities can be derived as a function of the choice of citizens to tax themselves (see Goldhaber, 1999 for an equilibrium model of public expenditures. This article also contains a useful review of the issue of endogeneity). James (1993) also uses a political economy model in explaining the determinants of public educational spending across countries.

⁹ Due to data limitations, the instrumentation is only used for equations that do not have prices or quality variables in them.

¹⁰ In previous work, James (1993) assumes a recursive system to deal with the endogeneity of the public sector equation. Martinez-Vazquez and Seaman (1985) use Two-stage least squares (TSLS) method but do not discuss identification issues.

private expenses — PTA, educational materials (books, school supplies), and miscellaneous expenses (uniforms, transportation). These cost variables are available for four years: 1986/87, 1989/90, 1992/93 and 1994/95 for 14 of the 16 regions in the sample.¹¹ Note that these cost variables are deflated by GDP deflators. Moreover, in order to control for the remaining effects of unobserved aggregated shocks, we also estimated models with year- and region-specific fixed effects.

Another variable used is GDP per capita, to measure average per capita provincial income for any given year. We also include overall provincial population for the following reasons. First, since we already have GDP per capita, doing so would capture the effects of aggregate income. Second, provinces with more people may simply tend to demand more spaces for their children because of, for example, income effect or scale economies. Third, some students may be over- or under-aged. Hence, the demand for public education is heavily influenced not only by the age-specific population but also by the rest of population. The National Economic and Development Authority (NEDA) is the source for all economic and population aggregates. GDP per capita is available for most regions and years for which enrollment data are available. Population data are only available for two census years: 1990/91 and 1995/96. The average annualized growth rates for these two years in each province are used to infer provincial growth rates for all years between 1986 and 1998. Since age-specific population data are also available only for the two census years, we estimated annual age-specific population by using linear interpolation and extrapolation with 1990/91 and 1995/96 data sets.

In the analysis described below, the unit of observation is the region for any given year. It is impossible to obtain data for all of the variables in Eq. (1) for all provinces and for all years from 1986 to 1998. The sample used in this paper thus changes, depending upon the availability of data.

4. Empirical analysis

4.1. Secondary education

Private and public secondary enrollment in the Philippines are positively correlated and the correlation coefficient between the two variables is 0.8. While private school enrollment has risen along with public school enrollment, this by itself cannot be interpreted as evi-

dence that there is no substitution between the two types of education. There are several important intervening variables that could explain this relationship. For example, both private and public school enrollment at the secondary level are likely to be positively correlated with the indicator of economic well-being in the region, represented by GDP per capita and population. Hence, it is important to hold constant for these variables. Moreover, in order to cope with this spurious correlation problem, we also estimate the model by the instrumental variable method, treating the public school enrollment as an endogenous variable.

Table 1 shows the means and standard deviations for the variables used in this paper. These are the average regional means for an average year. They show that for secondary education, average enrollment in the public sector is almost twice as large as that in the private sector. Average test scores for private exceed those for public schools.

Table 2 shows the estimation results of Eq. (1) for secondary schools. The determinants of private sector enrollment include public sector enrollment, population and regional GDP at constant prices. All variables are strong predictors in that they are significantly different from zero. When population and GDP are held constant, public and private sector enrollment are negatively associated, and significantly so.¹² While there is not a one-for-one trade-off, the results indicate that enrolling 10 more students in public secondary schools would be associated with three or four *fewer* students enrolling in private secondary schools, resulting in a net gain in enrollment of six or seven students.¹³

Another way to measure substitutability is by examining the responsiveness of enrollment to quality. Higher

¹² Our results are derived by using the Huber–White robust standard error method. To check heteroskedasticity, we performed the Cook–Weisberg test of heteroskedasticity. The test result suggests that there is no evidence of heteroskedasticity (Table 2).

¹³ In percentage terms (elasticities calculated at the means), this means that a 1% increase in public school enrollment is associated with a decline in private school enrollment of about 0.70%. Other studies have used per pupil public school expenditures as a measure of both quantity and quality trade-offs between private and public schools. The elasticity estimate here is slightly larger than the range reported in other studies. In a cross-country analysis, James (1993) finds that the elasticity of the share of private schools in total enrollment with respect to per pupil spending in the public sector is -0.59 . In an analysis of school districts in the state of New York, Goldhaber (1999) finds that, in models that correct for endogeneity, per pupil public spending has a negative and significant effect on the private school enrollment rate of the order of -0.01 . The elasticities are not strictly comparable because of the different measures of the public school variable (public per pupil spending versus public enrollment) and of the private enrollment variable.

¹¹ The region 13 (Caraga region) and the region 16 (Autonomous Region of Muslim Mindanao) are dropped from our sample since their 1990/91 GDP and population information is not available.

Table 1
Summary statistics

	Mean ^a (standard deviation)	Minimum	Maximum
<i>Secondary education</i>			
Priv enrollment	104,824.0 (73,853.8)	27,991	277,080
Pub enrollment	202,516.6 (102,429.3)	45,881	480,680
Pub NSAT scores	104.5 (10.7)	82.8	131
Priv NSAT scores	122.2 (13.3)	92.2	161.7
Priv tuition (pesos) ^c	165.3 (44.0)	103.7	298.6
Priv tuition (pesos) ^c	1596.75 (252.9)	930.4	2106.6
Pub other costs (pesos) ^c	2165.4 (436.1)	1587.6	3627.0
Priv other costs (pesos) ^c	3491.4 (686.6)	2358.2	5472.6
Age-specific population at age 13–16	407,265.0 (171,520.8)	100,125	974,545
Population	4,325,677.0 (2,272,174.0)	1,146,000	10,700,000
Constant GDP per capita ^c	9983.2 (5316.2)	4399	29,047
Sample size	164 ^b		
<i>Elementary education</i>			
Priv enrollment	53,131.9 (69,795.3)	3230	317,313
Pub enrollment	695,270.3 (273,682.5)	179,951	1,599,953
Age-specific population at age 7–12	686,846.4 (276,373.9)	178,977	1,485,060
Population	4,274,513.0 (2,238,143.0)	1,146,000	10,700,000
Constant GDP per capita	9900.9 (5288.1)	3866	29,047
Sample size	178		
<i>Tertiary education</i>			
Priv enrollment	96,891.7 (114,709.7)	17,405	567,737
Pub enrollment	26,626.1 (21,431.2)	1767	133,669
Age-specific population at age 17–24	694,088.0 (381,247.9)	182,349	1,605,326
Population	4,455,745.0 (2,435,450.0)	1,146,000	10,700,000
Constant GDP per capita	10,097.2 (5300.1)	4781	29047
Sample size	112		

^a These means and standard deviations are calculated per region per year.

^b These are means based on all non-missing observations for each variable. Thus, the number of observations will vary, depending on the sample size.

^c These figures are in real terms.

quality in private schools should be positively associated with private enrollment. If public and private schools are substitutes in terms of quality, public quality should be negatively associated with private enrollment.

As noted in the previous section, data on quality, measured in terms of outcomes, such as performance in test scores, as opposed to inputs, such as pupil–teacher ratios, are difficult to obtain. The standardized test scores at secondary levels (NSAT) are available separately for public and private schools only from 1995 onwards at the regional levels. Consequently, the sample size is reduced significantly when these variables are included in the estimation, as shown in specification (2) of Table 2.¹⁴

¹⁴ In order to test whether the results of specification (1) change when the sample sizes used in specifications (2) and (3) are used, we also estimated Eq. (1) using the smaller sample sizes. The results show that the coefficients for population and GDP at constant prices are not affected much by the variation in sample size. The sign and significance of the negative coefficient on public enrollment remain, while the size of the coef-

The results show that the coefficients are imprecisely estimated and are not significantly different from zero. It is possible that parents do not respond to test scores as a proxy for school quality. Another explanation is that the results simply reflect an endogenous relationship between public school quality and private school enrollment, which generates a multicollinearity problem. However, we cannot test this because of data limitations.

Even if quality variables are included in the regression, the public enrollment coefficient on private enrollment remains significantly negative. An increase of 10 students enrolled in public schools is associated with a drop in private school enrollment of more than two students. Greater public school enrollment, of a given quality, leads to lower private enrollment.

The classical definition of substitution revolves around the cross-price elasticity of demand. An increase in the

ficient changes somewhat. The results, which are not shown in this paper, are available from the authors upon request.

Table 2

Determinants of private school enrollment in Philippine secondary schools: estimation method: ordinary least squares with robust standard errors; dependent variable, private secondary school enrollment^a

Specification	(1)	(2)	(3)
Intercept	−36,608.272 (7.16)***	−13,940.575 (0.47)	−12,426.343 (0.53)
Pub enrollment	−0.362 (7.25)***	−0.282 (2.15)**	−0.312 (2.96)***
Age-specific population (13–16)	0.099 (3.03)***	0.125 (1.39)	0.029 (0.52)
Population	0.034 (9.78)***	0.027 (2.41)**	0.040 (7.37)***
Constant GDP per capita	2.651 (6.59)***	3.729 (2.03)**	2.601 (3.58)***
Pub NSAT scores		−21.350 (0.03)	
Priv NSAT scores		−273.053 (0.40)	
Pub tuition (pesos)			−140.627 (1.32)
Priv tuition (pesos)			−40.596 (1.80)*
Pub other costs (pesos)			33.067 (1.81)*
Priv other costs (pesos)			−2.035 (0.14)
Cook–Weisberg test statistics of heteroskedasticity	0.13	0.44	0.41
[P-value]	[0.72]	[0.51]	[0.52]
Observations	164	56	53
R ²	0.91	0.93	0.93
School years	1986–98	1995–98	1987, 1990, 1993, 1995

^a Huber–White robust *t*-statistics in parentheses. *, Significant at 10% level; **, significant at 5% level; ***, significant at 1% level. The data are for the end of each school year.

price of public education would be associated with a rise in private enrollment. Because of the limited number of observations in the price variables, the sample size is reduced to 53. Specification (3) in Table 2 shows that private tuition fees are significantly and negatively related to private enrollment, as might be expected. The own-price elasticity of demand (evaluated at the means) of private tuition is -0.62 .¹⁵

The cross-price elasticity of demand for private education is not significantly different from zero. The most plausible explanation for this is that the price variables may be capturing important quality characteristics. Greater average spending is correlated with higher quality, which would explain increased enrollment. This may counteract the price effect. One way out of this problem

is to include a quality variable in the regression, as was done in the specification (2) of Table 2. Unfortunately, test scores for private and public schools are not available for the subsample of regions that have price data, except for one year (1994–95). On the other hand, the effect of the non-tuition public cost variable on private enrollment is positive and statistically significant. This result suggests that there is a standard substitution effect in secondary education. More importantly, the key variable, public school enrollment, retains its negative and significant relationship with private school enrollment in the specification (3) of Table 2 with the imperfect price information. Indeed, the coefficient is almost identical to that found in the specification (1) of Table 2.

As noted in the previous section, it is possible that the public enrollment variable may be endogenous. Specification (4) of Table 3 shows the results of the basic equation when public enrollment is instrumented, using political elections as the identifying instrumental variable. The basic qualitative results do not change. The

¹⁵ This result should be interpreted with caution since it is likely that private tuition is also endogenous. Tuition is responsive to other variables. However, it is difficult to instrument for tuition given the present data base.

Table 3

Determinants of private school enrollment in Philippine secondary schools: estimation method, instrumental variable method with robust standard errors without and with year and regional fixed effects; dependent variable, private secondary school enrollment^a

Specification	Without fixed effects (4)	With fixed effects (5)
Intercept	−35,799.988 (5.87)***	12,719.422 (0.70)
Pub enrollment ^b	−0.404 (3.53)***	−0.535 (3.92)***
Age-specific population (13–16)	0.102 (3.18)***	0.143 (2.97)***
Population	0.036 (6.29)***	0.034 (9.14)***
Constant GDP per capita	2.600 (5.81)***	1.698 (0.97)
Observations	164	164
R ²	0.91	0.98
School years	1986–98	1986–98

^a Huber–White robust *t*-statistics are in parentheses. *, Significant at 10% level; **, significant at 5% level; ***, significant at 1% level. Estimated results of fixed effects are not reported here for an expositional purpose.

^b Represents endogenous variable.

coefficient of public enrollment becomes even more negative and remains significantly different from zero. It was not possible to use the instrumented public enrollment variable together with the quality and tuition variables because the reduced sample size made the instrument almost perfectly correlated with the constant. Finally, we estimated a model with year- and region-specific fixed effects. The result is reported as the specification (5) of Table 3.

The results indicate that basic qualitative results do not change even if we control for the potential omitted variable bias. In fact, the substitution effects are reinforced if we control for endogeneity and omitted variable bias. Enrollments of 10 more students in public secondary schools would decrease private secondary school enrollments by five students, indicating a very strong crowding-out effect.

4.2. Primary and tertiary education

The effect of public on private enrollment is different for primary and tertiary education (Table 4). In the case of primary education, the effect is not conclusive. The effect of public enrollment is still negative but not statistically significant as can be seen in specification (6) of Table 4. The absolute magnitude of the substitution

Table 4

Determinants of private school enrollment in Philippine elementary and tertiary schools: estimation method, instrumental variables regression with year and regional fixed effects; dependent variable: private elementary or tertiary school enrollment^a

Specification	Elementary (6)	Tertiary (7)
Intercept	−79,571.220 (3.25)***	−14,397.651 (0.18)
Pub enrollment (elementary level) ^b	−0.158 (0.92)	
Pub enrollment (tertiary level) ^b		1.265 (0.79)
Age-specific population (7–12)	0.028 (0.34)	
Age-specific population (17–24)		−0.453 (2.07)**
Population	0.048 (3.88)***	0.113 (2.05)**
GDP at constant prices	2.421 (1.18)	−29.264 (1.55)
Observations	178	112
R ²	0.98	0.88
School years	1986–98	1991–98

^a Huber–White robust *t*-statistics are in parentheses. *, Significant at 10% level; **, significant at 5% level; ***, significant at 1% level.

^b Represents endogenous variable.

coefficient is considerably smaller than that of secondary education: an increase of public school enrollment by 10 will lead to a decline in private enrollment by fewer than two. However, in elasticity terms evaluated at the average, the effect is −1.41. This is simply because the share of private school in total enrollment at the primary level is much smaller than that at the secondary level. Even though the private sector in Philippine primary education occupies a relatively small niche, enrollment there may also be affected by the size of the public sector. The Philippines has very high enrollment rates at primary school ages and, with everyone already participating, the substitution between public and private education is inevitable.

The situation in tertiary education is very different. Although there should be more overlap in tertiary education in terms of quality, increased public enrollment is probably able to draw from non-participants because coverage is still relatively low. In fact, public enrollment is positively related to private enrollment, although the effect is not statistically significant according to the instrumental variable result of the specification (7) in

Table 4.¹⁶ However, we cannot make assessments about cross-price elasticities since it was not possible to include quality and price variables in these models for elementary and tertiary education because of severe data limitations.

5. Conclusions

A heavier public sector involvement in education may be warranted for efficiency reasons in order to compensate for market failures due to externalities and the lack of an effective credit market. Public sector expansion should raise overall enrollment. In countries where there is little or no private sector involvement, we would expect this to be close to one-for-one. However, if there is already an active private sector, then some of the public sector expansion may be achieved by drawing away students who would have gone to school anyway — in the private sector. If so, then, the efficiency gains may be attenuated.

This analysis has shown that expanding public education is negatively associated with private sector enrollment at secondary levels. While the magnitude of the crowding out is not one-for-one, it is significant. Enrolling 10 more students in public secondary schools would be associated with three to five *fewer* students enrolling in private secondary schools, resulting in a net gain in enrollment of five to seven. This result is maintained, even after holding constant for the effects of public school quality and prices, the possible endogeneity of the public sector, and omitted variable bias. Indeed, when endogeneity is taken into account, the magnitude of the substitution effect becomes larger — 10 more public school students are associated with four or five fewer private school students. Thus, when assessing the success of the expansion in public secondary education, one should not assume that this is done in a vacuum. The private sector reacts and the net effect on enrollment may not be as large as one would expect.

The degree of substitutability in terms of direct quantities of schooling varies by level of education. At primary levels, with already high overall participation rates,

there may be significant private–public substitution in terms of quantity as well. On the other hand, at tertiary levels the trade-off is not as apparent, probably because public sector expansion there has succeeded partly through greater school participation rates.

Public intervention to raise quality or lower the price of public schools, on the other hand, may not affect private sector enrollment, holding constant for public sector enrollment. One might be tempted to conclude that in choosing its instruments of intervention, lowering public school prices or raising quality may have less detrimental effects on private enrollment than expanding quantity at secondary and even primary levels. However, severe data limitations preclude the rigorous testing of this conjecture.

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Appendix A

Table 5

Determinants of private school enrollment in Philippine tertiary schools: estimation method, instrumental variables regression with year and regional fixed effects, sample without the National Capital Region (Metro Manila); dependent variable, Private tertiary school enrollment^a

Specification	Tertiary (7')
Intercept	36,019.54 (0.687)
Pub enrollment (tertiary level) ^b	0.334 (0.375)
Age-specific population (17–24)	–0.206 (0.953)
Population	0.063 (1.333)
GDP at constant prices	–19.888 (1.481)
Observations	104
R ²	0.54
School years	1991–98

^a Huber–White robust *t*-statistics are in parentheses.

^b Represents endogenous variable.

¹⁶ As pointed out by a referee, the results for tertiary education may be different from those in other levels because students are more likely to migrate across regions than for lower levels. In order to investigate this issue, we estimate the tertiary education model by dropping the National Capital Region (Metro Manila) from our sample (Appendix A). The basic qualitative results are not sensitive to this sample change.

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